10) Doppler shift

Velocity = 100 km/h, carrier frequency = 900 MHz

Convert velocity:

$$v = \frac{100}{3.6} = 27.78 \text{ m/s}$$

$$f_d = \frac{vf}{c} = \frac{27.78 \times 900 \times 10^6}{3 \times 10^8}$$

$$= 83.33 \text{ Hz} \quad \text{(maximum Doppler shift)}$$